EXERCISE 1 Find the values of the real numbers a and b if the line

$$2x - y = 2$$

is a tangent to the curve

$$y = a \sin 3x + b \cos x$$

at $x = 0$.

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EXERCISE 2 A particle moves along the x-axis such that at time t seconds its position is x metres from O, where

$$x = \frac{1}{2} + t + \frac{1}{2}\cos 4t.$$

Find :

1. the number of times the particle comes to rest in the first 6 seconds ;
2. the distance travelled between the first two positions of rest ;
3. the position of the particle at the first instant of zero acceleration.